

# Review Summery

## RipFinder: real-time rip current detection on mobile devices

*For Internal Model Development Reference*

Khan et al., Frontiers in Marine Science, 2025

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### Key Contributions

- Real-time, on-device rip current detection using mobile-optimised ML models.
- Client-server hybrid architecture: on-device models run offline; a server runs heavier models when online.
- Detection of two visually distinct rip current types: bathymetry-controlled and transient rips.
- Citizen science data upload with automated and expert quality control.
- In-app educational content about rip current risks.

### System Architecture Summary

The app deploys up to two mobile-optimised models simultaneously on the device (EfficientDet-D2 + YOLOv8n for capable devices; one model for older hardware). When internet is available, images are also forwarded to a server running larger, more accurate models (YOLOv8l, RT-DETR-L). This redundancy reduces false negatives — a critical safety requirement.

### Model Performance

| Model           | Deployment | Avg. Accuracy | FPS (iPhone 12) | Notes                       |
|-----------------|------------|---------------|-----------------|-----------------------------|
| EfficientDet-D0 | Client     | 0.73          | 48              | Fast; misses transient rips |
| EfficientDet-D2 | Client     | 0.91          | 15              | Best mobile accuracy        |
| YOLOv8n         | Client     | 0.30          | 25              | Fast but low accuracy       |
| YOLOv8m         | Client     | 0.48          | 17              | Moderate balance            |
| RT-DETR-L       | Server     | 0.93          | N/A             | High accuracy server model  |
| RT-DETR-X       | Server     | 0.96          | N/A             | Best overall accuracy       |

- EfficientDet-D2 — the recommended client-side model. Best mobile accuracy at 0.91, still runs in real time. The clear choice for on-device deployment.
- RT-DETR-X — the recommended server-side model. Highest overall accuracy at 0.96 with the tightest confidence interval, making it the most reliable when internet is available

### Strengths

- Works offline — critical for remote beaches without connectivity.
- Dual-model strategy on the client reduces missed detections (false negatives).
- Separates two visually distinct rip classes rather than merging them into one.
- Statistical validation (95% CIs) adds credibility to model selection decisions.
- Citizen science loop enables continuous dataset growth and model improvement.

## **Weaknesses & Gaps to Address in Our Model**

- Training data skewed toward elevated (drone/satellite) viewpoints; ground-level beach perspective is underrepresented — a major real-world usability gap.
- YOLOv8 variants performed poorly (YOLOv8n avg 0.30), yet are still included as primary client models, raising reliability concerns.
- EfficientDet-D0 completely failed on one test video (accuracy = 0.00), suggesting fragility in edge cases.
- Dataset remains geographically limited despite stated diversity goals; only ~9,345 images total across both rip classes.
- No real-world user trial data presented — only lab video tests used for evaluation.
- Server dependency introduces latency and single point of failure; no offline fallback for high-accuracy needs.
- IoU threshold of 0.3 is lenient — our model should target stricter thresholds (e.g., 0.5+).

## **Recommendations for Improved Model**

- Prioritize beach-level training data to close the viewpoint gap identified in RipFinder.
- Use a stricter IoU threshold ( $\geq 0.5$ ) to better reflect real-world detection precision.
- Explore model architecture beyond YOLO/EfficientDet — e.g., transformer-based segmentation models for better boundary delineation of amorphous rip shapes.
- Include temporal context (short video clips / optical flow) rather than single-frame detection to capture the dynamic movement signature of rip currents.
- Invest in geographic dataset diversity from the outset (Australia, SE Asia, Mediterranean).
- Consider a lightweight on-device segmentation model rather than bounding-box detection for more precise rip area delineation.